



Armada Kalamunda Group

Hypertensive Disorders of Pregnancy Guideline

1. Purpose

To standardise the care of women/ Pregnant people with hypertension, preeclampsia, and eclampsia in pregnancy. To assist doctors and midwives in their decision making in the detection, treatment, and care of women with hypertension in pregnancy, and those who present with or develop preeclampsia and/or eclampsia. From here on all pregnant people will be referred to as women

2. Applicability

This guideline applies to medical and nursing staff providing care for woman at Armada Kalamunda Group (AKG).

3. Glossary

The following definitions are relevant to this policy.

Term	Definition
Gestational Hypertension	Gestational hypertension refers to hypertension with systolic BP ≥ 140 mmHg or Diastolic blood pressure ≥ 90 mmHg without proteinuria or other signs/symptoms of preeclampsia that develops after 20 weeks of gestation. BP should be elevated on two separate occasions at least six hours apart. (Detecting a rise in blood pressure from 'booking' or preconception blood pressure ($> 30/15$ mmHg) should lead to closer monitoring but is not diagnostic of hypertension in pregnancy.
Pre-eclampsia	Preeclampsia refers to the new onset of hypertension and proteinuria after 20 weeks of gestation in a previously normotensive woman. It is defined as systolic blood pressure of 140mmHg and/or diastolic blood pressure of 90mmHg or higher measured on at least two occasions over several hours, combined with proteinuria >300 mg total protein in a 24-h urine collection, or ratio of protein to creatinine >30 mg/mmol.

Term	Definition
Chronic Hypertension	Chronic/pre-existing hypertension is defined as systolic pressure ≥ 140 mmHg and/or diastolic pressure ≥ 90 mmHg that antedates pregnancy or is present before the 20th week of pregnancy (on at least two occasions) or persists longer than 12 weeks postpartum. It can be primary (essential hypertension) or secondary to a variety of medical disorders.
Eclampsia	Eclampsia is the development of grand mal seizures in women with preeclampsia

4. Guideline

4.1 Risk factors for gestational hypertension, pre-eclampsia and eclampsia

- Hypertensive disease/preeclampsia in a previous pregnancy
- Multiple pregnancy
- Pre-existing medical condition:
 - Hypertension
 - Type 1 or 2 diabetes
 - Antiphospholipid antibody syndrome or Systemic Lupus Erythematosus
 - Renal disease.
 - Vascular & connective tissue disorders
 - Thrombophilia's:
 - Antiphospholipid syndrome
 - Protein C and S deficiency
 - Antithrombin III deficiency and
 - Factor V Leiden
- Obesity, BMI $\Rightarrow 30$
- Maternal age <18 or >35
- Nulliparous
- Family history of preeclampsia
- New partner

4.2 Procedure for recording blood pressure in pregnancy

The systolic blood pressure is accepted as the first sound heard (K1) and the diastolic blood pressure the disappearance of sounds completely (K5). Where K5 is absent, K4 (muffling) should be accepted. Correct cuff size is important for accurate blood pressure recording. Manual sphygmomanometers remain the gold standard for measurement of blood pressure in pregnancy. Automated blood pressure recorders and aneroid sphygmomanometers may be used but are prone to error and should be calibrated on a regular basis.

White coat Hypertension: White Coat hypertension is defined as hypertension in a clinical setting with normal blood pressure away from this setting.

5. Gestational hypertension

5.1 Management

Degree of hypertension	Mild hypertension (140/90 to 149/99 mmHg)	Moderate hypertension (150/100 to 159/109 mmHg)	Severe hypertension (160/110 mmHg or higher)
Admit to hospital	No	No	Yes
Treat	No	With oral labetalol as first-line treatment to keep: diastolic blood pressure between 80–100 mmHg and systolic blood pressure less than 150 mmHg	With oral labetalol as first-line treatment to keep: diastolic blood pressure between 80–100 mmHg systolic blood pressure less than 150 mmHg
Measure Blood pressure	Not more than once a week	At least twice a week	At least four hourly
Test for protein urine	At each visit using automated reagent-strip reading device or urinary protein: creatinine ratio	At each visit using automated reagent-strip reading device or urinary protein: creatinine ratio	Daily using automated reagent-strip reading device or urinary protein: creatinine ratio
Blood tests	Only those for routine antenatal care	Test kidney function, electrolytes, full blood count, LFT, uric acid, LDH	Test at presentation and then monitor
CTG	If clinically indicated	Twice weekly or more if clinically indicated	Daily or more while inpatient if indicated
Ultrasound	US for fetal growth/MVP/UADoppler(UAD). At time of diagnosis and then weekly for MVP and UAD and fortnightly for fetal growth velocity depending on the severity of illness.		

5.2 Treatment

Women with gestational hypertension should have tight blood pressure control to a target of $\leq 130/85$ mmHg

Medication	Dose	Maximum dose in 24hours
Labetalol (avoid in women with asthma)	100mg-400mg bid-qid	1600mg maximum dose in 24hrs
Methyldopa	250mg-500mg bid-qid	3-4g
Nifedipine normal release	10mg-20mg bid-tds	120mg

Only offer women with gestational hypertension antihypertensive treatment other than labetalol after considering side-effect profiles for the woman, fetus and newborn baby. Alternatives include methyldopa and Nifedipine.

- In women receiving outpatient care for severe gestational hypertension, after it has been effectively controlled in hospital, measure blood pressure and test urine twice weekly and carry out weekly blood tests.
- In women with mild hypertension presenting before 32 weeks, or at high risk of pre-eclampsia, measure blood pressure and test urine twice weekly.
- Do not offer bed rest in hospital as a treatment for gestational hypertension unless the woman falls into the admission category.

5.3 Timing of delivery

It is not recommended to offer birth before 39 weeks to women with mild to moderate gestational hypertension, with or without antihypertensive treatment. Discuss with specialist Obstetrician about plan of management if delivery not contemplated.

For women with gestational hypertension whose blood pressure is lower than 160/110 mmHg after 37 weeks, with or without antihypertensive treatment, timing of birth, and maternal and fetal indications for birth should be agreed between the woman and the senior obstetrician.

Discuss risks and benefits of delivery with patient if preterm and delivery is required due to severity of hypertension. Offer intramuscular steroid injection for lung maturity if delivery expected within a week.

6. Pre-eclampsia

The basis of management is stabilisation, assessment, observation and if appropriate, delivery.

The frequency and place of surveillance is based upon the gestation, the severity and rate at which the condition advances.

- Mild preeclampsia in Outpatient setting
- Moderate to severe preeclampsia in hospital

Degree of hypertension	Mild hypertension (140/90 to 149/ 99 mmHg)	Moderate hypertension (150/100 to 159/ 109 mmHg)	Severe hypertension (160/110 mmHg or higher)
Admit to hospital	No	Yes	Yes

Degree of hypertension	Mild hypertension (140/90 to 149/99 mmHg)	Moderate hypertension (150/100 to 159/109 mmHg)	Severe hypertension (160/110 mmHg or higher)
Treat	No	With oral labetalol as first-line treatment to keep: diastolic blood pressure between 80–100 mmHg systolic blood pressure less than 150 mmHg	With oral Nifedipine As first-line treatment to keep: diastolic blood pressure between 80–100 mmHg systolic blood pressure less than 150 mmHg
Measure blood pressure	Weekly	At least four times a day	More than four times a day, depending on Clinical circumstances
Test for proteinuria	At each visit using automated reagent-strip reading device or urinary protein: creatinine ratio	Daily using automated reagent-strip reading device or urinary protein: creatinine ratio	Daily using automated reagent-strip reading device or urinary protein: creatinine ratio
Blood tests	Monitor using the following tests twice a week: kidney function, electrolytes, full blood count, LFT, uric acid, LDH	Monitor using the following tests three times a week: kidney function, electrolytes, full blood count, LFT, coagulation if low platelets, uric acid,	Monitor using the following tests three times a week: kidney function, electrolytes, full blood count, LFT, coagulation, LDH, uric acid
CTG	Weekly	Daily or more if indicated while hospitalised	Daily or more while inpatient if indicated

6.1 Treatment

Medication	Dose	Maximum dose in 24 hours
Labetalol BP 150/100-159/109 (avoid in women with asthma)	100mg-400mg bid-qid	1600mg
Nifedipine Slow Release (SR)BP 160/110	10mg-20mg bid-tds	120mg
Methyldopa	250mg-500mg bid-qid	3-4g

Betamethasone	11.4mg intramuscular injection daily for two doses 24 hours apart.
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Administer antenatal corticosteroids for lung maturation where appropriate. Recommended if <34 weeks gestation. (RCOG)

6.2 Severe pre-eclampsia

Is defined as systolic blood pressure 160 or higher and/or diastolic blood pressure of 110mmHg or higher measured on at least two occasions over several hours, combined with proteinuria >300 mg total protein in a 24-h urine collection, or ratio of protein to creatinine >30 mg/mmol in a spot urine sample. This is usually accompanied by symptoms, and/or neurological hyperactivity and can be accompanied by HELLP syndrome. **HELLP syndrome** is a variant of severe preeclampsia (**H**aemolysis, **E**levated **L**iver enzymes and **L**ow **P**latelet) count. HELLP may be an independent disorder. 15 to 20 percent of affected patients do not have concurrent hypertension or proteinuria.

Elements include:

- Thrombocytopenia (common)
- Haemolysis (rare) and
- Elevated liver enzymes (common)

This degree of hypertension requires urgent assessment and management.

Management of HELLP syndrome is similar to severe preeclampsia.

6.3 Sign and symptoms of system

Cardiovascular	• Severe blood pressure elevation • Systolic blood pressure ≥ 160 mm Hg or diastolic blood pressure ≥ 110 mm Hg on two occasions at least four hours apart
Renal	• Proteinuria: $\geq 1+$ on dipstick • Proteinuria confirmed by laboratory testing of a spot urine protein/creatinine ratio of ≥ 30mg/mmol • Oliguria i.e. <500mL/24 hours or <30mL/hour • Serum or plasma creatinine > 0.09mmol/L or 90μmol/L. • Rapid weight gain with or without generalised oedema. (Proteinuria is not mandatory to make the clinical diagnosis. Hyperuricaemia is a common but not diagnostic feature of pre-eclampsia but is an important predictor of maternal and fetal wellbeing)

Haematological	• Abnormal liver enzymes (ALT or AST rising to above 70 IU/litre). • Thrombocytopenia. platelet count < 100x10⁹ • Haemolysis as found with raised LDH and lowered Hb • Coagulation profile derangement (only taken if platelet count is low) • HELLP syndrome (Haemolysis, Elevated Liver enzymes, and Low Platelet count)
Hepatic	• Nausea and/or vomiting • Upper abdominal pain, often at the right upper quadrant, possibly due to liver capsule distention • Raised serum transaminase >70iu/L
Neurological	• Headache and/or visual disturbances (photopsia, scotomata, cortical blindness, retinal vasospasm) • Hyperreflexia with clonus • Convulsions (eclampsia) • Stroke
Respiratory	• Pulmonary oedema or cyanosis
Gestational	• Placental Abruption
Fetal	• Fetal growth restriction mostly Asymmetrical • Abnormal fetal heart rate on CTG • Oligohydramnios • Increased resistance, absent or reversed end diastolic flow on umbilical artery Doppler

6.4 Investigations

The minimum post-diagnostic laboratory/imaging evaluation should include:

- Full blood picture with particular reference to Platelet count
- Serum creatinine
- LDH
- Uric acid
- Serum aspartate aminotransferase (AST) or alanine aminotransferase (ALT)
- Urine Protein Creatinine ratio (PCR). Repeated urinary protein assessment in women with proteinuria from preeclampsia (in the absence of other indications) is not recommended. There is inadequate data to determine the severity of preeclampsia or timing of delivery based on urine protein assessment
- Obstetrical ultrasound for fetal wellbeing, liquor volume (MVP) and umbilical artery Doppler
- CTG

6.5 Treatment of severe pre-eclampsia

The optimum blood pressure range to achieve and maintain is: 140-160/90- 100mmHg. Blood pressure should be monitored continuously and reduced gradually to avoid adverse fetal side effects from a rapid decrease in uteroplacental perfusion. Continuous electronic fetal monitoring should be performed concurrently, until optimal blood pressures are maintained.

6.6 Medications

Labetalol		
Medication	Dose	Maximum dose
Intravenous Labetalol (To be avoided in women with asthma)	20mg given over 2 minutes. Repeat every 10 minutes	300mg
On-going Labetalol infusion via Infusion Pump Add 40mL (200mg) of Labetalol to 60mL of 0.9% sodium chloride. The resultant 100mL of solution contains 200mg Labetalol - 2mg/mL.)	20-160mg/hour titrated until optimal blood pressures are achieved. (10 – 80 ml/hr)	300 mg. Patients requiring ongoing management with Labetalol infusion should be discussed with ICU
Ongoing Labetalol infusion by syringe driver Draw up 200 mg (40 ml) undiluted (5 mg/ml)	Start infusion at 4mls/hr (20 mg/hr)	300 mg. Patients requiring ongoing management with Labetalol infusion should be discussed with ICU
Hydralazine		
Intravenous Hydralazine	5-10mg given over 5-10 minutes. Repeat every 20-30 minutes X 2	15mg
On-going Hydralazine infusion	5mg/hour to maintain optimal B/P	

7. Eclampsia

Refer to appendix 4: [Quick reference flowchart for Prophylaxis of Eclampsia](#)

Eclampsia is the development of grand mal seizures in women with preeclampsia and occurs in 1 in 200/300 women in Australia. It is unpredictable and often there are no associated clinical precursors. Seizures may occur in antenatal, intrapartum or postpartum period usually within 24 hours of birth but occasionally later. A differential diagnosis of eclampsia should be made until other medical, metabolic and neurological conditions have been excluded. The woman may or may not have signs or symptoms of preeclampsia.

7.1 Prophylaxis for eclampsia

Magnesium Sulphate (MgSO₄) is the drug of choice for eclampsia and is recommended for women as seizure prophylaxis in cases of severe preeclampsia. Although uncommon, Magnesium toxicity can occur, and Magnesium levels should be assessed if signs of toxicity are apparent. After birth the woman will be admitted to ICU.

MgSO₄ should be continued for at least 24 hours following birth or the last seizure, whichever is later.

- MgSO₄ comes as 8g in a 100ml bag

7.2 Medications

Medication	Dose	Maximum dose
MgSO ₄ Intravenous loading dose	4g given over 20 minutes. Set volume to be infused as 150ml /hrs over 20 mins on pump. Total infusion 50 mls (4 gms)	4g
MgSO ₄ Intravenous maintenance infusion	1g/hour. Set pump to 12.5mls per hour	1g/hr over 24 hours
MgSO ₄ Intravenous for recurrent seizures	2g given over 5 min Set VTBI at 150 mls/hr over 10 mins on pump. Total infusion 25 mls	2g

If seizure continues

Medication	Dose	Maximum dose
Diazepam	5 – 10 mg IV	10mg

7.3 Monitoring during Mg SO₄ infusion

- Every 15 minutely blood pressure, pulse, respiratory rate in the acute phase and then every 30 minutes
- 1 hourly patellar reflexes and oxygen saturation
- 1 hourly urine output measurement
- 2 hourly temperatures
- Continuous electronic fetal monitoring.
- ECG monitoring should be considered

7.4 MgSO₄ signs of toxicity

While magnesium toxicity is relatively uncommon, women with renal impairment are at a greater risk of toxicity while receiving MgSO₄ infusion.

- Suppression or loss of patellar reflexes
- Respiratory depression
- Drowsiness
- Loss of consciousness
- Abnormal Urea and electrolytes indicating kidney function impairment
- To treat toxicity, stop the infusion and give Calcium Gluconate 1 g (10 ml) IV over 10 minutes. Take Magnesium levels to confirm that altered state is due to MgSO₄ and not another cause.

7.5 Indications for delivery in severe pre-eclampsia

- Fetal Gestational age $\geq$ 37 weeks gestation
- Severe fetal growth restriction
- Inability to control hypertension
- Non-reassuring fetal status
- Deteriorating platelet count
- Deteriorating liver function
- Deteriorating renal function
- Placental abruption
- Persistent neurological symptoms or eclampsia
- Persistent epigastric pain, nausea or vomiting with abnormal liver function tests
- Acute pulmonary oedema

Severe preeclampsia is an indication for delivery regardless of gestation. However in mild or moderate preeclampsia, conservative management should be considered and particularly so if <34 weeks gestation.

- If 34–36 weeks gestation with non-severe preeclampsia, there is insufficient evidence to make a recommendation about the benefits or risks of expectant management.
- If $\geq$ 37 weeks, women with gestational hypertension or preeclampsia – expediting birth should be considered after discussion with the woman.

8. Birth

At Armadale we care for women who are 35 weeks and above gestation. However, in an emergency situation where a woman is too unstable to transfer we will care for women under 35 weeks' gestation and call the NETS team to transfer the baby to a level 3 nursery.

- If $\geq$ 34 weeks gestation, vaginal birth is optimal, if possible.
- If <32 weeks gestation, the success of induction is reduced, and a caesarean section may be necessary.

It is necessary to stabilize the woman and is important to discuss with the paediatrician and the anaesthetist before any plans for delivery at AHS are made. Keep ICU informed if patient has been started on Magnesium Sulphate.

Active third stage of labour is recommended using syntocinon as per protocol. Avoid Syntometrine.

Graduated compression stockings should be considered for all inpatients, with or without prophylactic low molecular weight heparin

8.1 Post-partum care

In cases of preeclampsia and eclampsia, blood pressure can rise in the first 3-6 postpartum days before it begins to normalise. It can however take several weeks before the blood pressure returns to normal. The incidence of postpartum eclampsia should not be underestimated, 44% of all cases of eclampsia occur in the postpartum period.

MgSO₄ should continue for a minimum of 24 hours following birth or after the last seizure whichever is later. Admission to ICU after birth is required for the duration of MgSO₄ administration for continued intensive one to one care.

- Antihypertensive should be titrated down.
- Maintain blood pressure < 160/110mmHg
- Inpatient care is prudent till the managing team is confident of patients' recovery

8.2 Follow up

- All women who have required intravenous drug therapy for Preeclampsia are to remain in hospital till stable and BP under control. Each day the Specialist Obstetrician will review the woman and plan discharge and follow up appropriately.
- Appropriate and timely communication with the community health provider, who should continue to monitor blood pressures and manage antihypertensive therapy.
- Severe preeclampsia 6-week obstetric review.
- Refer to appropriate specialist if there is persistent hypertension or proteinuria.
- Preconception counselling for subsequent pregnancies.

9. Chronic hypertension in pregnancy

Refer to [appendix 2](#): Quick reference flowchart: Chronic Hypertension

9.1 Management

- The woman with chronic hypertension should be observed frequently during pregnancy by an obstetrician and by physician familiar with management of hypertension in pregnancy.
- Treatment of chronic hypertension is associated with a significant reduction in severe hypertension but has not been shown to alter the risk of superimposed preeclampsia, preterm birth, placental abruption or perinatal death.

Treatment should be started when the blood pressure consistently reaches or exceeds 160 mmHg systolic and/or 100 mmHg diastolic.

- Treatment at BP levels between 140 and 159 mmHg systolic and/or 90–99 mmHg diastolic is also common practice, with good, documented outcomes such as damage secondary to chronic hypertension (for example, kidney disease)
- Women with gestational hypertension should have tight blood pressure control to a target of $\leq 130/85$ mmHg
- Do not offer pregnant women with uncomplicated chronic hypertension treatment to lower diastolic blood pressure below 80 mmHg.
- The drugs used for treatment of chronic hypertension are the same as those recommended for preeclampsia and gestational hypertension. Use LABETALOL first
- Encourage women with chronic hypertension to keep their dietary sodium intake low, either by reducing or substituting sodium salt, because this can reduce blood pressure.
- Offer pregnant women with secondary chronic hypertension referral to a specialist in hypertensive disorders.
- Fetal surveillance in women with chronic hypertension in pregnancy should include an early dating ultrasound in the first trimester and then fetal growth/MVP/UAD as per clinical assessment.

It is not recommended to offer birth to women with chronic hypertension whose blood pressure is lower than 160/110 mmHg, with or without antihypertensive treatment, before 37 weeks.

- For women with chronic hypertension whose blood pressure is lower than 160/110 mmHg after 37 weeks, with or without antihypertensive treatment, timing of birth and maternal and fetal indications for birth should be agreed between the woman and the senior obstetrician

9.2 Postnatal investigations, monitoring and treatment

In women with chronic hypertension who have given birth, measure blood pressure

- In women with chronic hypertension who have given birth, aim to keep blood pressure lower than 140/90 mmHg.
- In women with chronic hypertension who have given birth: continue antenatal antihypertensive treatment.
- Review long-term antihypertensive treatment 2 week after the birth.
- Restart the antihypertensive treatment the woman was taking before she planned the pregnancy.
- Women with chronic hypertension require a medical review at their postnatal review (6–8 weeks after the birth) with their community GP. Their own GP needs to be fully informed of the woman's condition, care and outcome during her stay in hospital on the discharge summary.
- Inform women who have had gestational hypertension or preeclampsia, and their primary care clinicians, that these conditions are associated with an increased risk of developing high blood pressure and its complications in later life.

10. General management in severe hypertension

10.1 Fluid management

Pulmonary oedema from excessive fluid administration is a major cause of maternal morbidity and mortality in cases of severe preeclampsia/eclampsia. A strict limit to fluid intake should be adhered to and input/output carefully documented.

- Strict fluid balance charting and recording on MAADS chart
- Restrict total fluid intake to 80mL/hr, if woman is dehydrated, increase the fluid intake, encourage oral intake
- In severe preeclampsia/eclampsia – use indwelling urinary catheter with hourly urometer to monitor output
- Observe for oliguria – if <30mL/hour for 2 or more hours, request specialist obstetrician/ GP obstetrician review
- Observe for pulmonary oedema

10.2 Anaesthetic considerations in hypertensive disorders of pregnancy

- Inform the duty anaesthetist about a woman with severe pre-eclampsia well prior to labour or operative delivery.
- For labour and delivery, epidural analgesia is a useful adjunct to antihypertensive therapy for blood pressure control and it improves renal and uteroplacental blood flow.
- Regional anaesthesia is preferred to general anaesthesia for caesarean section, especially as airway problems including laryngeal oedema may be increased.
- Women who develop organ failure require intensive monitoring and medical management and will be transferred to a tertiary centre.
- Regional analgesia/analgesia is appropriate in the absence of a coagulopathy.
- Preloading of fluids is not advised.
- Pre referral to an anaesthetist is recommended

11. Prophylaxis in prevention of severe hypertension

11.1 Low dose aspirin

- Low dose aspirin 150mg has moderate benefits when used for prevention of pre-eclampsia and its consequences and is safe to use in pregnancy. In view of this potential benefit, and the relative absence of maternal or neonatal complications, low dose aspirin is indicated for the secondary prevention of pre-eclampsia in women at increased risk. In most cases, aspirin may be ceased at 36 weeks gestation although continuation beyond this period is not unsafe.
- Advise pregnant women at high risk of pre-eclampsia to take 150 mg of aspirin daily at bedtime before 16 weeks of gestation until at least 36 weeks of gestation. Women at high risk are those with any of the following:
 - hypertensive disease during a previous pregnancy
 - chronic kidney disease
 - autoimmune disease such as systemic lupus erythematosus or antiphospholipid syndrome

- type 1 or type 2 diabetes
- chronic hypertension.
- Advise pregnant women with more than 1 moderate risk factor for pre-eclampsia to take 100-150 mg of aspirin daily from before 16 weeks until 36 weeks gestation. Factors indicating moderate risk are:
 - first pregnancy
 - age 40 years or older
 - pregnancy interval of more than 10 years
 - body mass index (BMI) of 35 kg/m² or more at first visit
 - family history of pre-eclampsia
 - multi-fetal pregnancy.

11.2 Supplemental calcium

The supplemental use of Calcium is strongly recommended in pregnant women with low dietary calcium intake <1gram/day. Consider assessing serum corrected calcium prior to commencement of calcium oral supplementation (to ensure the absence of underlying hypercalcaemia)

12. Long term consequences of hypertensive disorders of pregnancy

Women who have had preeclampsia in their pregnancy may be at a higher risk of the following later in life:

- 4x higher risk of developing high BP
- 2x higher risk of developing heart disease
- 2-4x higher risk of type 2 diabetes mellitus
- 2x higher risk of developing stroke
- 4-8x higher risk of developing kidney disease
- 2 in 3 women will die from a cardiac disease

13. Lower above risks post-partum

A history of Preeclampsia doesn't have to mean you will develop cardiovascular problems. Postpartum make changes and:

- Get regular exercise
- Eat a well-balanced, healthy diet
- Maintain healthy weight
- Stop smoking
- See your doctor for a regular health check
- Speak with your doctor before next pregnancy
- Take prescribed medications

14. Supporting information

The following documents support the implementation of this policy:

- Health Department of Western Australia (HDWA) [Clinical Handover Policy MP 0095](#)

- HDWA [Consent to Treatment Policy MP 0175/22](#)
- [Medication Management Procedure AKG-PRO-0255-18](#)
- [Recognition and Response to Acute Deterioration AKG-POL-0388-14](#)

15. Compliance, monitoring and evaluation

Compliance with this policy will be monitored via:

- Monitoring of reported clinical incident data via Datix CIMS.

Specific findings are to be reported at the relevant departmental meeting for review and identification of any opportunities for improvement.

16. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

17. References

1. SOMANZ, Hypertension in Pregnancy Guideline 2023
2. [3centres](#) Collaboration. Hypertension in Pregnancy, Preeclampsia and Eclampsia. Clinical Practice Guidelines Printed June 2010.
3. NHMRC Recommended guideline: Queensland Maternity and Neonatal Clinical Guideline: Hypertensive disorders of pregnancy: Document Number MN10.13-V4- R15
4. LOWE S. Guidelines for the management of hypertensive disorders of pregnancy., Australian and New Zealand Journal of Obstetrics and Gynaecology; 2009. 49:242– 246
5. Koopmans CM, Bijlenga D, Groen H et al. Induction of labour versus expectant monitoring for gestational hypertension or mild pre-eclampsia after 36 weeks' gestation (HYPITAT): a multicentre, open-label randomised controlled trial. The Lancet, Volume 374, Issue 9694, Pages 979 - 988, 19 September 2009
6. NICE Guideline, Hypertension in Pregnancy: Diagnosis and management: updated 2021
7. Clinical practice guideline, KEMH 2021

18. Document owner

Head of Obstetrics and Gynaecology

Enquiries relating to this guideline may be directed to:

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19. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V3	19/04/2024	19/04/2027	

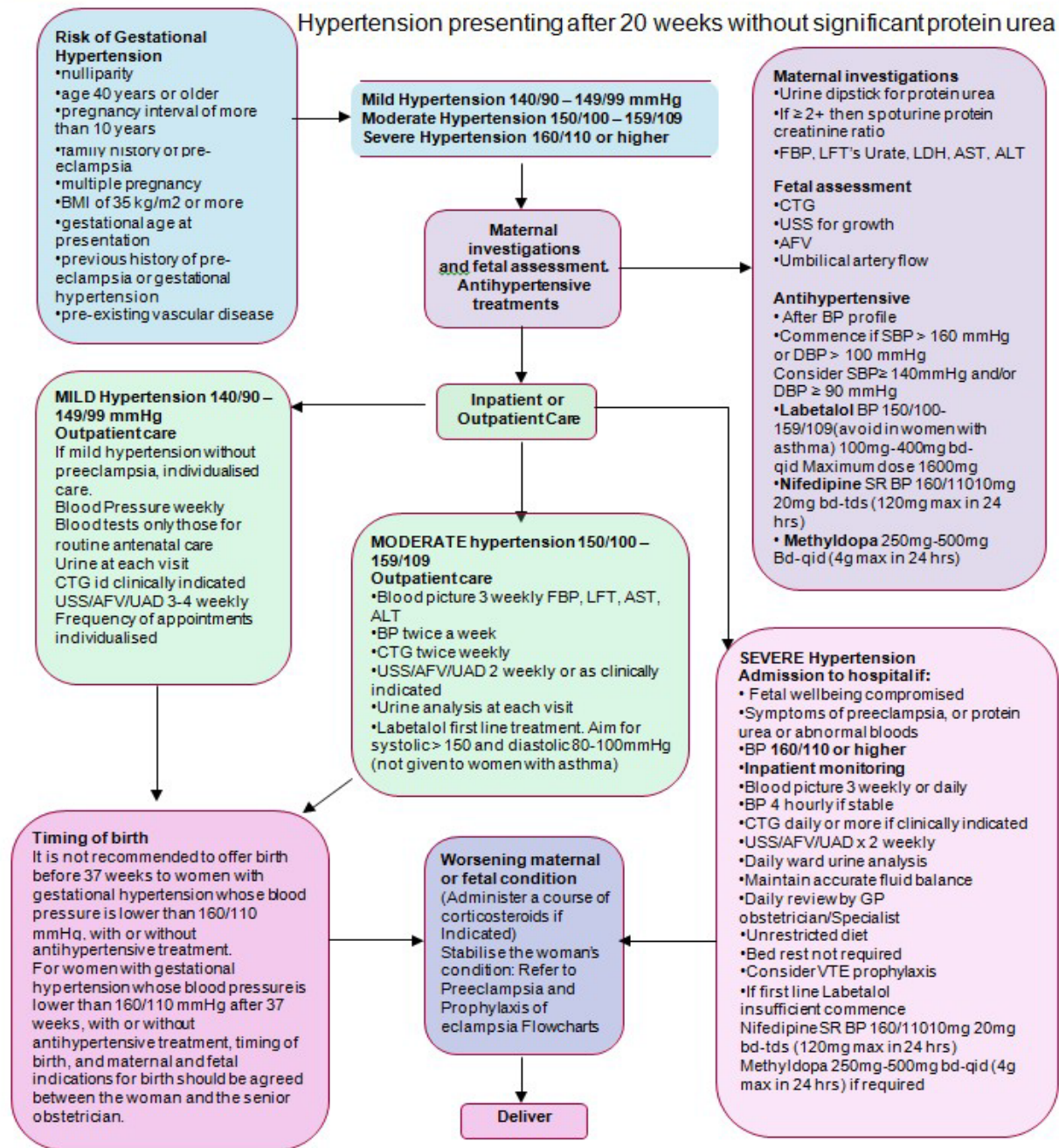
20. Authorisation

This guideline has been authorised and issued by the [enter position title of the authoriser].

Authorised by	Tania Strickland – A/Director Clinical Services
Authorisation date	19/04/2024
Published date	22/04/2024

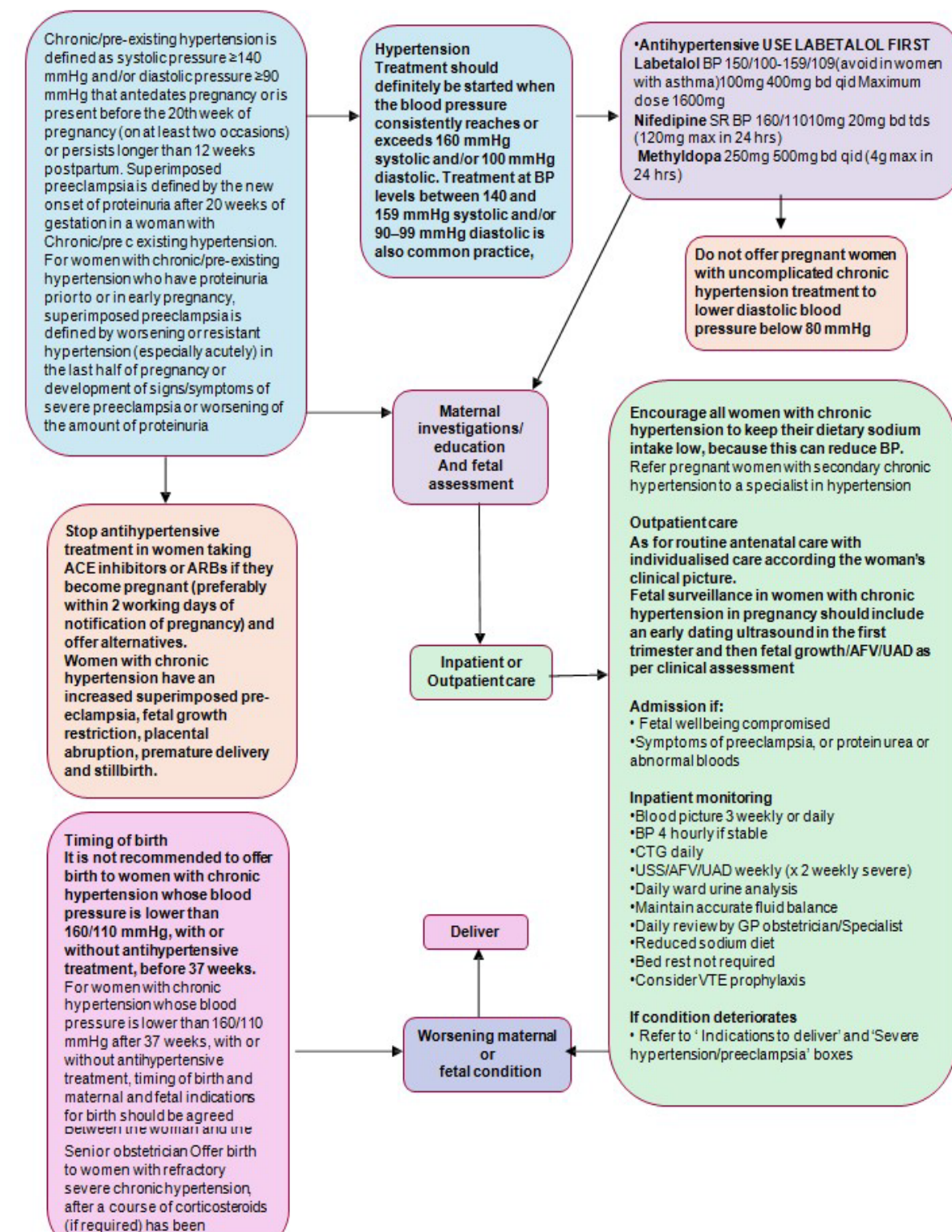
Appendix 1: Gestational hypertension

Hypertensive disorders of pregnancy guideline AHS Maternity

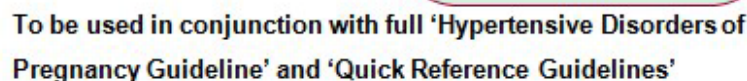


Appendix 2: Chronic hypertension flowchart

AHS Maternity Chronic Hypertension Flowchart



AHS Maternity Preeclampsia Flowchart



Appendix 4: Prophylaxis of eclampsia

AHS Maternity Prophylaxis of Eclampsia Flowchart

